



The three op-amps (A1 through A3) have similar circuits wired around them, differing only in the value of feedback resistors ($R1$, $R8+R13$, $R12$). The feedback resistors help in setting the gain of each

of the circuit.

The most striking feature of this circuit is dual voltage supply of $\pm 3V$. This has been done to ease circuit design by reducing the number of components. The power supply is built around a battery pack comprising four AA-size cells. These four cells are placed in series, with the positive terminal of the first cell fed to +3V line, the negative terminal of the last cell fed to -3V line, and the connection between the second and third

cells serving as 0V or ground line. Capacitors C7 and C8 (each 220 μ F) between +3V and 0V, and 0V and -3V, improve the AC performance of the circuit, by AC shorting the rails to ground potential.

The LM324, a popular quad op-amp IC, is used here. It introduces minimal noise and houses all the four op-amps in a single 14 dual-in-line package, thereby allowing for circuit compactness. Of the four op-amps, only three are used here.

stage, which is necessary to realise the Butterworth filter. Resistors R3 through R6, R9 and R10 in conjunction with capacitors C1 through C6 determine the low-pass cut-off frequency of the filter. ($f=1/(2\pi RC)=1/(2\pi \times 0.01\mu F \times 4.7k\Omega)=3.386$ kHz.

Assemble the circuit on a general-purpose PCB and enclose in a small box. You can bring out the input and output terminals on the front panel of box for ease of giving input and taking the output signal. ●

BEDROOM LIGHT



■ T.A. BABU

This circuit allows you enough time to reach your bed and lie down before the bedroom lamp switches off automatically. You can find a number of applications for this circuit. The circuit draws almost no power when it is inactive.

Mains power supply is reduced by the capacitive dropper arrangement. A 1F capacitor (C2) handles the timing part. When switch S1 is pushed to 'on' position for a few minutes, bulk capacitor C2 charges and the stored energy drives the internal LED of IC MOC3041

